
Software Requirements Specification

for

AcademiCore Instrument Allocation System

Prepared by Group 01

Group Members	Index
U.G.K.A. Madhubhashi	2023t01880
M.K.K.K. Sandaruwani	2023t01900
M.W.D.U.Vidwanthi	2023t01906
W.M.S.H. Bandara	2023t01851
S.W.A.R.Prathibha	2023t01889
P.R.Wijethunga	2023t01912

Supervisor : Dr.Rohan Samarasinghe

Department of Information and Communication Technology
Faculty of Technology
University of Colombo
22/23 batch

Table of Contents

Table of Contents	2
1. INTRODUCTION.....	1
1.1. Purpose	1
1.2. Document Conventions	1
1.3. Intended Audience and Reading Suggestions	2
1.4. Project Scope	2
1.5. References	2
2. OVERALL DESCRIPTION	3
2.1. Project Perspective	3
2.2. Project Functions	3
2.2.1. Student Functions.....	3
2.2.2. Lab Staff / Technical Officers.....	4
2.2.3. Administrator Functions	4
2.3. User Classes and Characteristics	4
2.4. Operating Environment	5
2.4.1. Hardware Platform.....	5
2.4.2. Operating System and Versions.....	5
2.4.3. Software Components.....	5
2.5. Design and Implementation Constraints	6
2.6. User Documentation	6
2.7. Assumptions and Dependencies	6
2.7.1. Assumptions.....	6
2.7.2. Dependencies	7
3. EXTERNAL INTERFACES AND REQUIREMENTS	8
3.1. User Interface	8
3.1.1. General Requirements.....	8
3.1.2. Main Components.....	8
4. SYSTEM REQUIREMENTS	9

4.1. Functional Requirements	9
4.2. Nonfunctional Requirements	11
5. SYSTEM SCENARIOS	13
5.1. Use-case Diagrams	13
5.2. Scenarios	14
5.2.1. SCN-1: Login	14
5.2.2. SCN-2: View Instruments	15
5.2.3. SCN-3: Reserve Instrument	16
5.2.4. SCN-4: Cancel Booking	18
5.2.5. SCN-5: Booking Approval (Optional)	19
5.2.6. SCN-6: Manage Instrument Status	20
5.2.7. SCN-7: Manage Inventory	21
6. OTHER REQUIREMENTS	22

1. INTRODUCTION

This document includes specifications and features of the software in detail. It helps understand the target audience and user classes accordingly and internal and external interface requirements.

1.1. Purpose

The purpose of this document is to outline the software requirements for the development of the Instrument Allocation System (IAS) designed for university students and staff across the ICT, IAT, ET, and AT departments.

This system aims to digitize and streamline the process of allocating laboratory and workshop instruments to students for academic purposes. Traditionally, the booking and management of instruments has been handled manually, leading to issues such as double-booking, lack of transparency, inefficient tracking, and misuse of resources. By automating this process, the university can significantly improve operational efficiency, traceability, and accessibility.

The key objectives of the system are:

- To provide a centralized web-based platform for students to search, view, and reserve instruments based on availability and department.
- To enable real-time updates on instrument status and booking history for all users.
- To empower staff and administrators to manage inventory, approve bookings (if required), and maintain system integrity.
- To eliminate manual paperwork, reduce human error, and enhance the accountability of instrument usage.
- To ensure fair and transparent usage of institutional resources among all students.

This document serves as a formal agreement between stakeholders (students, staff, and developers) regarding the expectations, features, and technical requirements of the system. It will also be used by quality assurance teams to design relevant test cases and by developers to implement the specified functionalities accurately.

1.2. Document Conventions

This document follows a structured and standardized format based on IEEE 830 guidelines for Software Requirements Specifications. The following conventions are used throughout the document to ensure clarity, consistency, and ease of understanding:

- Terminology:
 - User refers to any student, staff member, or admin accessing the system.
 - Booking refers to the process of reserving an instrument.
 - Instrument refers to lab or workshop equipment made available for academic use.
- Requirement Labels:
 - Functional requirements are labeled as **FREQ-XXX**, e.g., **FREQ-001: User Login**.
 - Non-functional requirements are labeled as **NFREQ-XXX**, e.g., **NFREQ-002: Usability**.
 - Use-case scenarios are labeled as **SCN-XXX**, e.g., **SCN-003: Book Instrument**.
- Text Formatting:
 - Bold text** is used to emphasize section titles, user roles, or important system components.
 - Italic text* is used for system features, states, or specific interface elements.

1.3. Intended Audience and Reading Suggestions

This Software Requirements Specification (SRS) document is intended for all stakeholders involved in the development and use of the Instrument Registration System (IRS) for the ICT, IAT, ET, and AT departments. The primary audience includes students, who will use the system to view and reserve instruments; laboratory staff or technical officers, who will manage reservations and monitor usage; and system administrators, who will oversee user access, manage inventory, and enforce booking policies. Additionally, the software development team will rely on this document to implement features accurately, while the quality assurance (QA) team will use it to prepare and validate test cases. Students and staff should focus on the functional requirements, user interface, and use-case scenarios. Developers should pay close attention to the detailed functional and non-functional requirements, system architecture, and software interfaces. QA engineers should refer to the requirements and scenarios to ensure the system meets all specifications. Overall, this document provides a shared understanding of the system's objectives and behaviors, serving as a reference throughout the development lifecycle.

1.4. Project Scope

The Instrument Allocation System (IAS) is being developed to serve as a centralized, web-based platform that enables students from the ICT, IAT, ET, and AT departments to efficiently browse, reserve, and manage access to laboratory and workshop instruments required for academic activities. The system aims to eliminate the current manual or semi-manual methods of instrument booking, which often lead to scheduling conflicts, underutilization, and administrative overhead. Through the IAS, students will be able to log in using their university credentials, view available instruments categorized by department and type, and submit booking requests based on availability. Lab staff or technical officers will have tools to monitor, approve, or reject these bookings (if applicable), and update instrument statuses such as “in use,” “available,” or “under maintenance.” Administrators will have full control over the system, including user and inventory management, access permissions, and data reporting. By automating the registration and booking process, the system ensures transparency, accountability, efficient resource utilization, and enhanced coordination between students and faculty. It also lays the foundation for future scalability and integration with other academic resource management systems.

1.5. References

The following documents, standards, and resources were referred to during the preparation of this Software Requirements Specification:

- IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications
- University Lab Equipment Usage and Booking Policy Manual (Internal Document)
- Developer Documentation for MySQL – for database schema planning
- [Optional] Reference SRS Documents from similar university systems (e.g., Library Management, Lab Reservation Tools)

2. OVERALL DESCRIPTION

2.1. Project Perspective

The Instrument Allocation System (IAS) is an internal web-based solution developed to improve the management and allocation of academic instruments within the university. It is designed to replace the current manual or spreadsheet-based methods used by departments such as ICT, IAT, ET, and AT. The IAS will operate as a client-server system accessible via desktop and mobile browsers, ensuring that both students and staff can interact with it conveniently from any location. The system will include secure authentication, a reservation management module, and administrative control panels. It is expected to integrate seamlessly with the university's existing infrastructure, including student databases and email authentication systems.

2.2. Project Functions

The Instrument Allocation System (IAS) is designed with distinct functions tailored to the three main user types: Students, Lab Staff, and Administrators. Each user class will access the system with role-based permissions that control what functions are available to them.

2.2.1. Student Functions

Students are the primary users of the system. Their responsibilities are focused on reserving and managing instrument usage for academic purposes.

- **Account Access**
 - Log in using university credentials
 - View and update profile
- **Instrument Browsing & Booking**
 - View available instruments by department or category
 - Filter/search instruments by availability, type, or usage purpose
 - View instrument details (name, type, status, availability time slots)
- **Booking Management**
 - Reserve an instrument for a specific date and time
 - View upcoming and past bookings
 - Cancel bookings (based on policy)
 - Receive booking confirmations via email

2.2.2. Lab Staff / Technical Officers

Lab staff are responsible for monitoring instrument usage, ensuring fair allocation, and managing the instrument lifecycle.

- **Booking Oversight**
 - View all incoming booking requests
 - Approve or reject bookings (if departmental approval is enabled)
 - View booking history for auditing
- **Instrument Management**
 - Mark instruments as “in use,” “available,” “damaged,” or “under maintenance”
 - Add usage comments or notes for specific instruments
 - Generate department-specific usage logs
- **Communication**
 - Notify students about booking changes or instrument issues via system messages or email

2.2.3. Administrator Functions

Administrators manage the system's users, policies, and inventory. They have the highest level of access.

- **User Management**
 - Add/remove users and assign roles
 - Reset passwords or update account information
- **Inventory Control**
 - Add new instruments to the system
 - Remove or archive retired/broken instruments
 - Edit instrument metadata (e.g., type, department, availability status)
- **System Monitoring**
 - View system-wide booking analytics and reports
 - Monitor department-level activity
 - Configure booking policies (e.g., max booking time, overlap restrictions)
- **Audit and Security**
 - Access logs of system actions (e.g., booking changes, user login history)
 - Perform regular backups and ensure data protection policies are enforced

2.3. User Classes and Characteristics

There are three primary user roles in the system: Students, Lab Staff/Technicians, and Administrators.

- Students are expected to have basic computer literacy and will use the system primarily for booking and viewing instruments.
- Lab Staff or Technical Officers are moderately technical users responsible for approving bookings, updating instrument statuses, and managing scheduling conflicts.
- Administrators are typically IT or department heads who manage the system configuration, control access permissions, and ensure that instruments are allocated fairly and used according to institutional policy.

2.4. Operating Environment

The Instrument Allocation System (IAS) is a web-based application designed to be accessible from various platforms and devices within the university network and beyond. It follows a client-server architecture, where users interact with the system through a web interface while the backend processes data and manages database operations. The system is built to support responsive design for seamless access on desktops, laptops, tablets, and smartphone.

2.4.1. Hardware Platform

- **Client Side (Users – Students, Staff, Admins):**
 - Devices: Desktop computers, laptops, smartphones, or tablets
 - Minimum Requirements: 2 GB RAM, 1 GHz dual-core processor, modern web browser
- **Server Side (University-hosted or Cloud-hosted Server):**
 - CPU: Quad-core or higher
 - RAM: 8 GB minimum
 - Storage: Minimum 200 GB (scalable based on user activity)
 - Internet: Reliable high-speed connection for handling simultaneous requests

2.4.2. Operating System and Versions

- **Client Devices:**
 - Compatible with Windows (10 or higher), macOS, Linux, Android (9.0+), and iOS (13+)
- **Server:**
 - Linux-based distributions recommended (e.g., Ubuntu Server 22.04 LTS)
 - Compatible with Windows Server 2019+ if needed

2.4.3. Software Components

- **Frontend:**
 - Web technologies: HTML, CSS, JavaScript
 - Responsive design frameworks: Bootstrap
- **Backend:**
 - Programming Languages: PHP (Laravel)
 - RESTful API architecture for frontend-backend communication
- **Database:**
 - Relational Database Management System (RDBMS): MySQL
 - ORM tools: Eloquent (for Laravel)
- **Authentication:**
 - OAuth 2.0 or custom login using university email credentials
 - Integration with the university's Single Sign-On (SSO) system (if applicable)
- **Web Server:**
 - Apache HTTP Server for hosting the web application

- **Notifications:**
 - SMTP server for email notifications (e.g., Gmail SMTP, university mail server)
 - Optional: Integration with Firebase Cloud Messaging for push notifications
- **Security:**
 - HTTPS enforced using SSL/TLS certificates
 - Role-based access control for all users
 - Input validation and data encryption mechanisms

2.5. Design and Implementation Constraints

- The system must enforce login via verified university email accounts only.
- Bookings should be limited to valid university students only and must comply with departmental instrument usage policies.
- Security measures such as HTTPS, secure password storage, and input validation must be implemented.
- Mobile responsiveness is required to support students using tablets or smartphones.
- The system must be modular to allow future integration with RFID-based tracking or university ERP systems.

2.6. User Documentation

User manuals and training guides will be provided in PDF format and published on the university intranet. These will include step-by-step instructions, frequently asked questions, and screenshots to assist students, staff, and administrators in using the system effectively. Helpdesk contact details will also be included for technical support.

2.7. Assumptions and Dependencies

It is assumed that all users (students, staff, and administrators) have access to internet-connected devices and university email accounts. The system's functionality depends on consistent access to the university's authentication system and database server. It also relies on cooperation from each department to maintain an accurate list of available instruments and approve bookings in a timely manner.

2.7.1. Assumptions

- **Reliable Internet Connection**

It is assumed that all users (students, staff, and administrators) have access to a stable internet connection to interact with the system.

- **Device Availability**

Users are expected to access the system via internet-enabled devices such as computers, smartphones, or tablets.

- **User Training and Awareness**

Students and staff are assumed to have basic technical knowledge and will receive instructions or training to use the system effectively.

- **Instrument Data is Accurate and Updated**

Each department will maintain and update its instrument inventory regularly to ensure accurate availability status.

- **Booking Rules are Predefined**

Booking rules (e.g., max duration, cancelation limits) will be set and agreed upon before system deployment.

- **Access to University Email System**

All users will have active university email accounts for login and notifications.

- **Institutional Support is Available**

IT and administrative staff will be available to maintain and support the system after deployment.

2.7.2. Dependencies

- **Authentication System Integration**

The IAS depends on integration with the university's Single Sign-On (SSO) or email-based authentication service.

- **Database Server Availability**

The system relies on a continuously running and well-maintained database server to store booking and instrument records.

- **Email Service (SMTP)**

Email communication for notifications and confirmations depends on a functioning SMTP email service or university mail server.

- **Instrument Inventory Accuracy**

The system's usefulness depends on departments regularly updating the instrument status (e.g., available, in use, under maintenance).

- **Compliance with University IT Policies**

The system must align with the university's data protection, privacy, and IT security guidelines.

- **Third-Party Library/Tool Support**

Any frameworks, APIs, or libraries used (e.g., Bootstrap, Firebase, REST APIs) must be stable and supported.

- **User Cooperation**

The success of the system depends on users following proper booking etiquette and institutional policies.

3. EXTERNAL INTERFACES AND REQUIREMENTS

3.1. User Interface

The User Interface (UI) section describes the look and feel of the system, how users will interact with it, and the components involved.

3.1.1. General Requirements

- **Responsiveness:**

The system interface must adapt to various screen sizes (desktops, laptops, tablets, smartphones).

- **Accessibility:**

The UI must follow accessibility guidelines (such as WCAG 2.1) to ensure usability for students or staff with disabilities.

- **Browser Compatibility:**

The system should work across major web browsers including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

- **Consistency:**

Navigation, colors, typography, and buttons should be uniform across all pages to ensure a consistent user experience.

- **Minimal User Steps:**

Common tasks like booking or canceling an instrument should require no more than 3–4 steps

3.1.2. Main Components

- **Login/Registration Screen:**

- Fields: Email, password
- Options: “Forgot Password”, “Reset Password”
- Login through university email credentials only

- **Student Dashboard:**

- View available instruments
- Search/filter by department or instrument type
- Reserve instrument and view booking history
- Cancel upcoming reservations

- **Staff Dashboard:**

- View pending and approved bookings
- Approve/reject bookings
- Update instrument status (available, in use, under maintenance)

- **Admin Dashboard:**

- Manage user roles and permissions
- Add/edit/delete instruments
- Generate usage reports and manage system settings

4. SYSTEM REQUIREMENTS

System requirements for a instrument Allocation system(IAS) encompass both functional and non-functional requirements necessary to design, develop, and implement the system effectively. These requirements define what the system should do, how it should perform, and the conditions under which it must operate

4.1. Functional Requirements

The functional requirements define the core operations that the Instrument Allocation System (IAS) must support to fulfill its purpose. Students must be able to register or log in using their university email credentials and gain access based on predefined roles such as Student, Lab Staff, or Administrator. The system should allow students to view a catalog of available instruments by department or category and reserve them for specific dates and time slots. Once reserved, students should be able to review their booking history and cancel upcoming reservations if permitted. Staff members will have the ability to monitor and manage incoming reservation requests, update the status of instruments (e.g., available, in use, under maintenance), and approve or deny booking requests when required. Administrators will manage the instrument inventory and user accounts, assign user roles, and maintain institutional policies through the system. Additionally, the system should automatically send email notifications for booking confirmations, updates, and cancellations. A reporting feature should be available for administrators to generate usage statistics and track instrument allocation across departments

FREQ reference	System Feature	Description and Priority
FREQ 001	User Registration & Login	Allows students, staff, and admins to log in using university email. (High Priority)
FREQ 002	Role-Based Access Control	Provides access based on user roles: Student, Staff, or Admin. (High Priority)
FREQ 003	View Instruments	Students can view instruments filtered by department, availability, and category. (High)
FREQ 004	Reserve Instrument	Students can select a date, time slot, and reserve available instruments. (High)
FREQ 005	Cancel Booking	Students can cancel upcoming bookings under allowed conditions. (Medium Priority)

FREQ 006	Booking Approval (Optional)	Staff can approve/reject bookings based on departmental rules. (Medium Priority)
FREQ 007	Update Instrument Status	Staff can update status of instruments (in use, under maintenance, etc.). (High Priority)
FREQ 008	Manage Instrument Inventory	Admins can add, edit, or remove instruments from the system. (High Priority)
FREQ-009	User Management	Admins can manage users and assign roles. (Medium Priority)
FREQ-010	Booking History & Tracking	Users can view past and current bookings with statuses. (Medium Priority)
FREQ-011	Email Notifications	System sends emails for confirmations, updates, and alerts. (High Priority)
FREQ-012	Generate Reports	Admins can generate usage reports by department or instrument. (Low Priority)

4.2. Nonfunctional Requirements

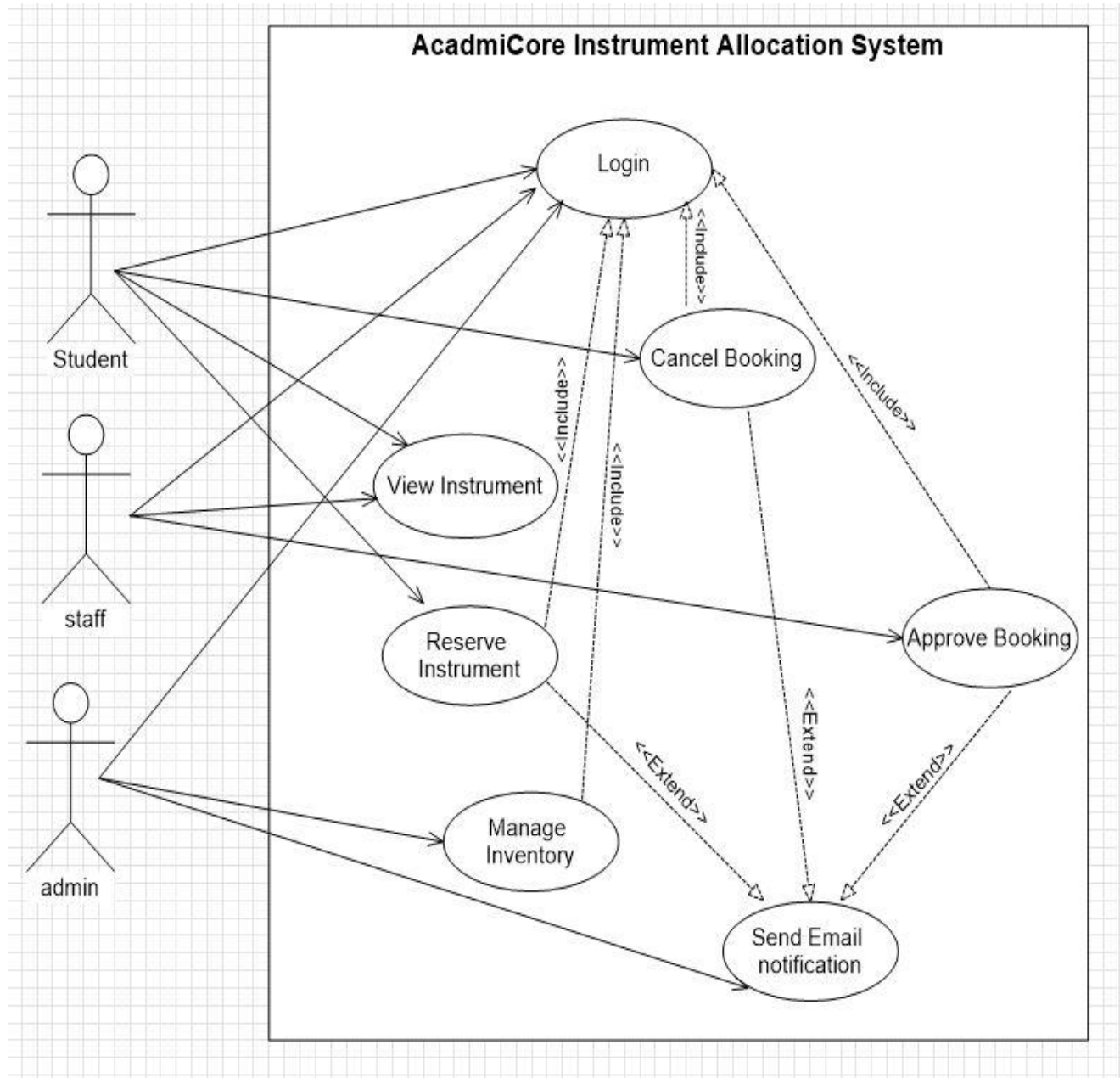
The non-functional requirements ensure that the system is usable, reliable, secure, and scalable to meet the operational needs of the university. The IAS must deliver high performance, ensuring fast response times even during peak usage by multiple users. Usability is critical, so the interface must be user-friendly and accessible, allowing students and staff with minimal technical expertise to use it effectively. The system must maintain high availability with minimal downtime and provide secure access through encrypted communication channels and safe data storage. Maintainability is also essential; the system should be modular and structured in a way that allows future enhancements or bug fixes without major rewrites. The system should be scalable, capable of supporting additional departments, instruments, or users as the university expands. It must also work consistently across various devices and browsers, ensuring accessibility for users on mobile, tablet, or desktop platforms. Data integrity must be upheld to prevent booking conflicts and ensure accurate inventory tracking. Finally, regular automated backups and disaster recovery mechanisms must be in place to prevent data loss and ensure business continuity.

NFREQ reference	System Feature	Description and Priority
NFREQ 001	Performance	System should respond within 2 seconds under normal load. (High Priority)
NFREQ 002	Usability	UI must be clean, responsive, and easy to navigate. (High Priority)
NFREQ 003	Reliability	99.5% uptime is expected with minimal downtime. (High Priority)
NFREQ 004	Security	All data (login, bookings) must be encrypted in transit and at rest. (High Priority)

NFREQ 005	Maintainability	Code should follow modular structure for easy updates. (Medium Priority)
NFREQ 006	Scalability	System should handle growth in users, instruments, and departments. (Medium Priority)
NFREQ-007	Compatibility	Must work on all major browsers and devices. (High Priority)
NFREQ-008	Data Integrity	Prevent duplicate/overlapping bookings. Ensure accuracy. (High Priority)
NFREQ-009	Backup and Recovery	System must perform daily backups and offer data recovery. (Medium Priority)

5. SYSTEM SCENARIOS

5.1. Use-case Diagrams



Assumption: To access all function actor must login first

5.2. Scenarios

This section describes key interaction flows between users and the Instrument Allocation System (IAS) in the form of real-world scenarios. Each scenario outlines how specific users (students, staff, or administrators) engage with the system to perform tasks such as logging in, viewing or reserving instruments, managing bookings, and administering inventory. These scenarios serve as detailed behavioral models that illustrate both the expected system responses and user actions under defined conditions.

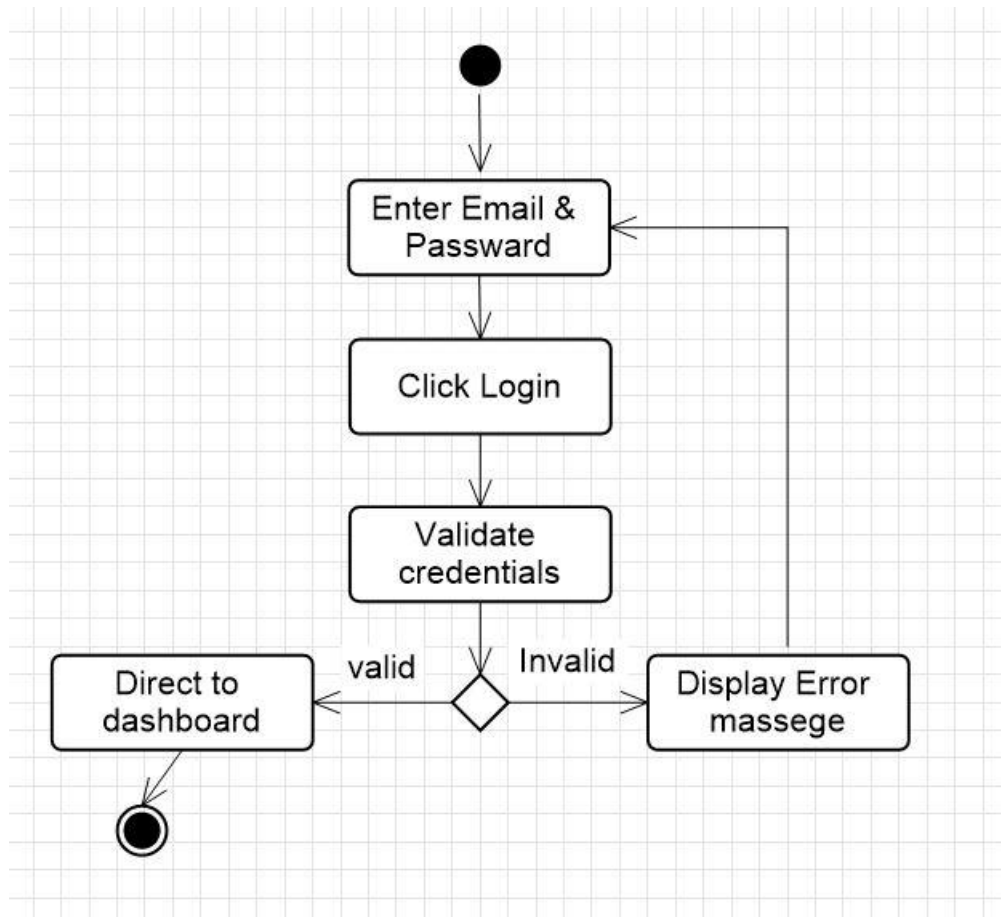
Each scenario is linked to relevant functional and non-functional requirements and includes clearly defined preconditions, basic flows, and postconditions. These scenarios also form the foundation for system testing, as they help developers and quality assurance teams understand the flow of data and logic across various modules of the system. The scenarios have been selected to reflect the core functions most frequently used by each user type (Student, Staff, and Admin) and are presented in tabular format for clarity and traceability.

5.2.1. SCN-1: Login

FREQ reference	FREQ-001 : Authentication and Authorization
NFREQ reference	NFREQ-001 (Performance)
Short Description:	This scenario describes how a user (student/staff/admin) logs into the system using university email.
Precondition:	The user has an active university email and an internet connection.

Basic flow: Login		
Step	User action	System response
1	Enters email and password	System validates the credentials
2	Clicks "Login"	System displays main menu if credentials are valid
3	If invalid credentials	Displays error message
Post-condition:		The user is authenticated and redirected to their appropriate dashboard.

Scenario Diagram for SCN-1 Login

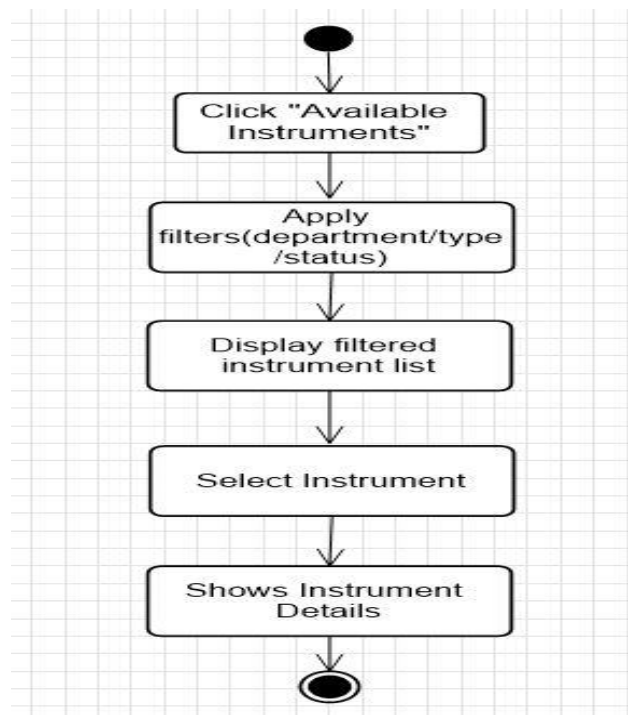


5.2.2. SCN-2: View Instruments

FREQ reference	FREQ-003 : View Instruments
NFREQ reference	NFREQ-002 : Usability
Short Description:	Students browse and filter available instruments by department, type, and availability.
Precondition:	The student is logged into the system.

Basic flow: View Instruments		
Step	User action	System response
1	Clicks on "Available instruments"	retrieves instrument list
2	Applies filters	Displays filtered results
3	Clicks on an instrument	Displays detailed information
Post-condition:		Student can view detailed and updated availability of instruments.

Scenario Diagram for SCN-2 View Instruments

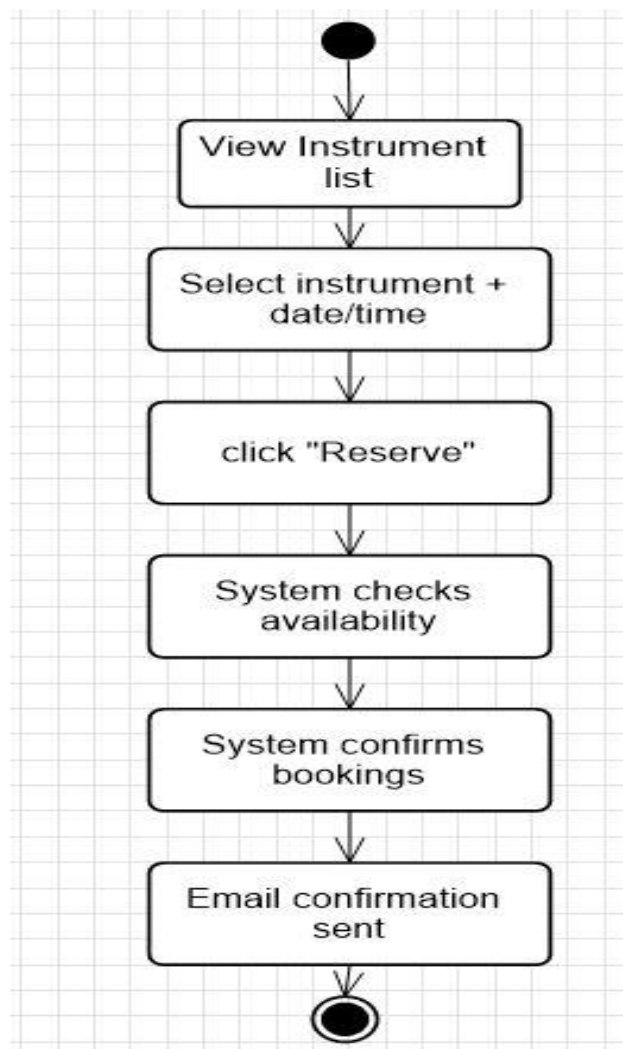


5.2.3. SCN-3: Reserve Instrument

FREQ reference	FREQ-004 : Reserve Instrument
NFREQ reference	NFREQ-001 ,00NFREQ 003
Short Description:	Student reserves an instrument by selecting time/date.
Precondition:	The student is logged in and has viewed the instrument list.

Basic flow: Reserve Instrument		
Step	User action	System response
1	Selects instrument and time slot	Validates availability
2	Clicks "Reserve"	Creates booking and sends confirmation
Post-condition:		Reservation is saved and confirmation email is sent.

Scenario Diagram for SCN-3 Reserve Instrument

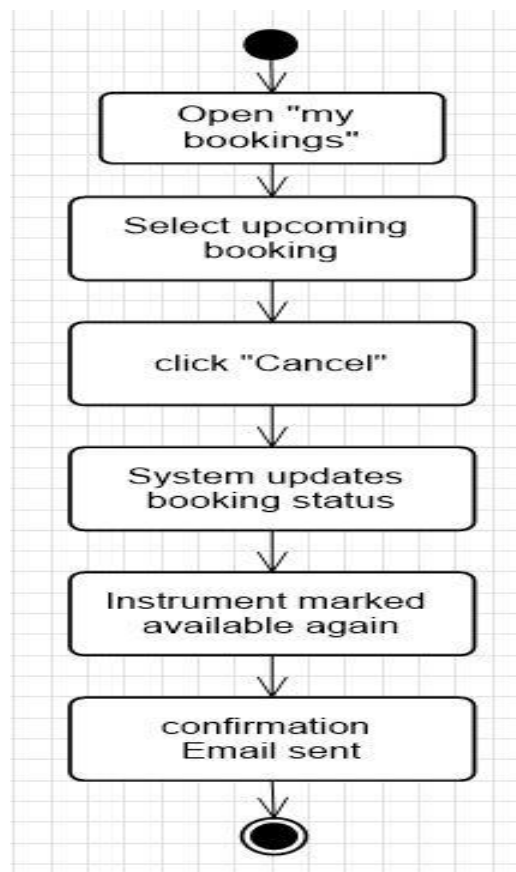


5.2.4. SCN-4: Cancel Booking

FREQ reference	FREQ-005 Cancel Booking
NFREQ reference	NFREQ-002 : Usability
Short Description:	Student cancels a previously made booking..
Precondition:	The student has a valid future booking

Basic flow: Cancel Booking		
Step	User action	System response
1	Navigates to “My Bookings”	Shows list of bookings
2	Clicks “Cancel”	Deletes booking and updates instrument status
3	Receives confirmation email	Notifies student
Post-condition:		The instrument becomes available again.

Scenario Diagram for SCN-4 Cancel Booking

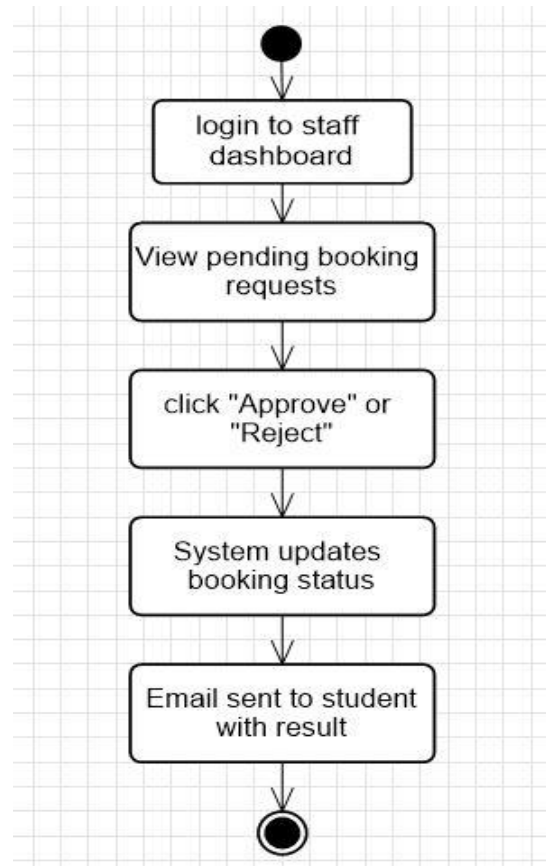


5.2.5. SCN-5: Booking Approval (Optional)

FREQ reference	FREQ-006 : Booking Approval
NFREQ reference	NFREQ-003 : Reliability
Short Description:	Staff approves or rejects a student's booking request
Precondition:	A new booking request is pending.

Basic flow: Booking Approval		
Step	User action	System response
1	Staff logs into dashboard	Displays pending requests
2	Clicks "Approve" or "Reject"	Updates booking status and sends email
Post-condition:		Student is notified of the booking decision

Scenario Diagram for SCN-5 Booking Approval

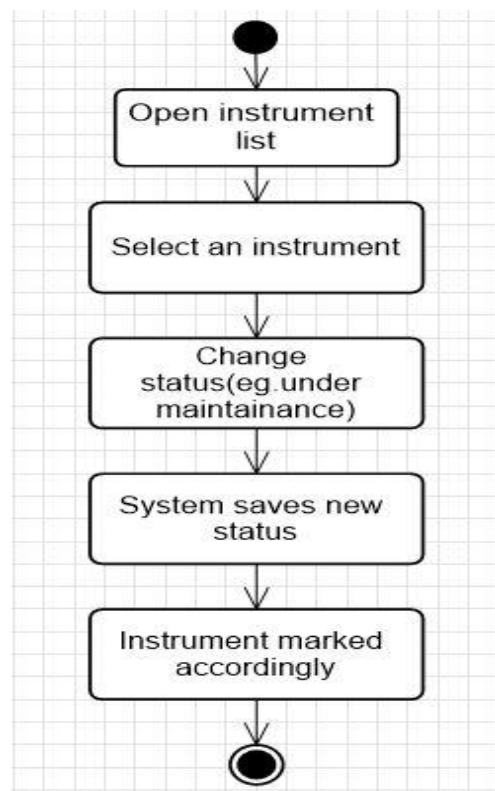


5.2.6. SCN-6: Manage Instrument Status

FREQ reference	FREQ-007 : Update Instrument Status
NFREQ reference	NFREQ-008 : Data Integrity
Short Description:	Staff updates instrument condition (e.g., in use, maintenance).
Precondition:	Staff is logged in.

Basic flow: Manage Instrument Status		
Step	User action	System response
1	Selects an instrument	Displays current status
2	Changes status to new value	Saves update and logs action
Post-condition:		Instrument status is updated in the system.

Scenario Diagram for SCN 6- Manage Instrument Status

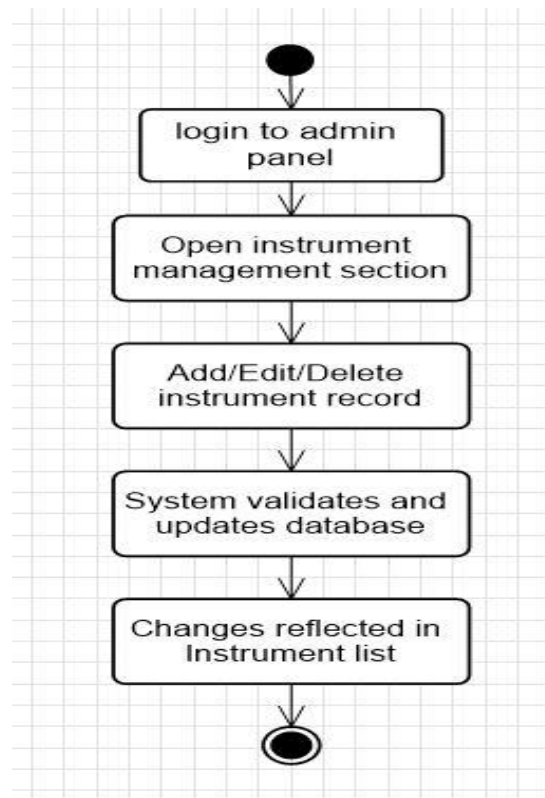


5.2.7. SCN-7: Manage Inventory

FREQ reference	FREQ-008 : Manage Inventory
NFREQ reference	NFREQ-005 : Maintainability
Short Description:	Admin adds, edits, or deletes instruments.
Precondition:	Admin is logged in.

Basic flow: Manage Inventory		
Step	User action	System response
1	Opens Inventory Manager	Loads current instruments
2	Adds or edits item	Saves changes
3	Deletes item (if needed)	Removes from list
Post-condition:		Inventory database is updated accordingly.

Scenario Diagram for SCN 7- Manage Inventory



6. OTHER REQUIREMENTS

- Cost
 - It is desirable to aim for a system with a minimum cost subject to the condition that it must satisfy all the requirements.
- Database Requirement
 - The system should use a relational database management system (RDBMS) like MySQL or PostgreSQL. Database schema should be normalized to eliminate redundancy and ensure data integrity.
- Reuse Objectives
 - The system should be designed with a modular architecture to facilitate the reuse of components across different projects or modules.